

An Analysis of Decision tree

Decision tree

Decision tree -- Part 1

The decision tree illustrates that when sequentially distributing lifeguards, placing a first lifeguard on beach #1 would be optimal if there is only the budget for 1 lifeguard. But if there is a budget for two guards, then placing both on beach #2 would prevent more overall drownings.

Decision tree -- Part 2

Decision tree using flowchart symbols Commonly a decision tree is drawn using flowchart symbols as it is easier for many to read and understand. Note there is a conceptual error in the "Proceed" calculation of the tree shown below; the error relates to the calculation of "costs" awarded in a legal action.

Decision tree -- Part 3

We will set D, which is the depth of the decision tree we are building, to three ($D = 3$). We also have the following data set of cancer and non-cancer samples and the mutation features that the samples either have or do not have. If a sample has a feature mutation then the sample is positive for that mutation, and it will be represented by one. If a sample does not have a feature mutation then the sample is negative for that mutation, and it will be represented by zero. To summarize, C stands for cancer and NC stands for non-cancer. The letter M stands for mutation, and if a sample has a particular mutation it will show up in the table as a one and otherwise zero.

Decision tree -- Part 4

Decision trees can also be seen as generative models of induction rules from empirical data. An optimal decision tree is then defined as a tree that accounts for most of the data, while minimizing the number of levels (or "questions"). Several algorithms to generate such optimal trees have been devised, such as ID3/4/5, CLS, ASSISTANT, and CART.

Decision tree -- Part 5

Now, we can use the formulas to calculate the phi function values and information gain values for each M in the dataset. Once all the values are calculated the tree can be produced. The first thing to be done is to select the root node. In information gain and the phi function we consider the optimal split to be the mutation that produces the highest value for information gain or the phi function. Now assume that M1 has the highest phi function value and M4 has the highest information gain value. The M1 mutation will be the root of our phi function tree and M4 will be the root of our information gain tree. You can observe the root nodes below

Decision tree -- Part 6

Increasing the number of levels of the tree The accuracy of the decision tree can change based on the depth of the decision tree. In many cases, the tree's leaves are pure nodes. When a node is pure, it means that all the data in that node belongs to a single class. For example, if the classes in the data set are Cancer and Non-Cancer a leaf node would be considered pure when all the sample data in a leaf node is part of only one class, either cancer or non-cancer. A deeper tree is not always better when optimizing the decision tree. A deeper tree can influence the runtime in a negative way. If a certain classification algorithm is being used, then a deeper tree could mean the runtime of this classification algorithm is significantly slower. There is also the possibility that the actual algorithm building the decision tree will get significantly slower as the tree gets deeper. If the tree-building algorithm being used splits pure nodes, then a decrease in the overall accuracy of the tree classifier could be experienced. Occasionally, going deeper in the tree can cause an accuracy decrease in general, so it is very important to test modifying the depth of the decision tree and selecting the depth that produces the best results. To summarize, observe the points below, we will define the number D as the depth of the tree. Possible advantages of increasing the number D:

Decision tree -- Part 7

Now, once we have chosen the root node we can split the samples into two groups based on whether a sample is positive or negative for the root node mutation. The groups will be called group A and group B. For example, if we use M1 to split the samples in the root node we get NC2 and C2 samples in group A and the rest of the samples NC4, NC3, NC1, C1 in group B. Disregarding the mutation chosen for the root node, proceed to place the next best features that have the highest values for information gain or the phi function in the left or right child nodes of the decision tree. Once we choose the root node and the two child nodes for the tree of depth = 3 we can just add the leaves. The leaves will represent the final classification decision the model has produced based on the mutations a sample either has or does not have. The left tree is the decision tree we obtain from using information gain to split the nodes and the right tree is what we obtain from using the phi function to split the nodes.

Decision tree -- Part 8

Runtime issues Decrease in accuracy in general Pure node splits while going deeper can cause issues. The ability to test the differences in classification results when changing D is imperative. We must be able to easily change and test the variables that could affect the accuracy and reliability of the

An Analysis of Decision tree

Decision tree

decision tree-model.

Decision tree -- Part 9

Drawn from left to right, a decision tree has only burst nodes (splitting paths) but no sink nodes (converging paths). So used manually they can grow very big and are then often hard to draw fully by hand. Traditionally, decision trees have been created manually – as the aside example shows – although increasingly, specialized software is employed.

Decision tree -- Part 10

They are unstable, meaning that a small change in the data can lead to a large change in the structure of the optimal decision tree. They are often relatively inaccurate. Many other predictors perform better with similar data. This can be remedied by replacing a single decision tree with a random forest of decision trees, but a random forest is not as easy to interpret as a single decision tree. For data including categorical variables with different numbers of levels, information gain in decision trees is biased in favor of those attributes with more levels. Calculations can get very complex, particularly if many values are uncertain and/or if many outcomes are linked.